

Circuit and working

The system can be divided into three sections: the power supply, audio, and LM3914 display-driver sections.

The power supply section is built around X1, BR1, and IC1. The 230V AC mains input is stepped down to 12V AC using transformer X1. This 12V AC is converted into pulsating DC by the full-wave bridge rectifier BR1. The rectified output is filtered by electrolytic capacitor C1 to reduce ripple before feeding the LM7805 regulator (IC1). IC1 provides a regulated 5V DC supply to power the system.

The unregulated 12V supply drives the LEDs (LED1 through LED40). LED41 serves as the power indicator, and resistor R1 limits current through it.

The audio section is built around audio jack J1, PNP transistor T1, and several passive components. The audio signal is applied through J1 and coupled to the circuit via resistor R4 and capacitor C3. Capacitor C3 blocks any DC component and passes only the AC component of the signal.

Switch S1 provides mode selection. When the common pin (pin 2) is connected to pin 1, the system operates in bar mode. When connected to pin 3, it operates in dot mode.

Transistor T1 amplifies the audio signal and feeds it to pin 5 of the LM3914 as the input. Capacitor C2 filters and smooths the amplified signal before it reaches the driver IC.

The LED display driver section is built around the LM3914 LED driver IC and the associated LEDs. The LM3914 is a monolithic display driver that senses the input voltage level and drives LEDs arranged on a linear scale. Internally, the IC compares the input signal with ten equally spaced reference levels derived from its built-in reference network.

Individual current-limiting resistors R5 through R14 (100Ω) are connected in series with each group of four LEDs to ensure uniform brightness and protect the LEDs.

Operation of the system is straightforward. As the audio signal amplitude increases, the output pins L1 through L10 of IC2 are enabled (go low) based on the signal level, illuminating the connected LEDs sequentially.

When configured in bar mode, all LEDs up to the detected level remain illuminated, forming a continuous bar display that follows the audio intensity.

Construction and testing

An actual-size, single-sided PCB layout for the audio signal-controlled LED display system is shown in Fig. 3, while the corresponding component layout is shown in Fig. 4. After assembling the circuit on the PCB, mount it

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